FIN2020 Homework 4 **Due by 23:59 March 16, 2026 (Monday)**

Please put your section number on your homework.

1. The Gains From Diversification

Consider portfolios formed from two risky assets, the first with expected return equal to $\mu_1 = 10$ and standard deviation of its return equal to $\sigma_1 = 8$ and the second with expected return equal to $\mu_2 = 6$ and standard deviation of its return equal to $\sigma_2 = 6$. Let w denote the fraction of wealth in the portfolio allocated to asset 1 and $1 - w$ the corresponding fraction of wealth allocated to asset 2.

a. Calculate the expected return and the standard deviation of the return on the portfolio that sets $w = 1/2$, assuming that the correlation between the two returns is $\rho_{12} = 0$.

b. Calculate the expected return and the standard deviation of the return on the same portfolio that sets $w = 1/2$, assuming instead that the correlation between the two returns is $\rho_{12} = -0.50$.

c. For which of these two values of ρ_{12} are the gains from diversification larger?

2. Portfolio Allocation with Mean-Variance Utility

Consider an investor with preferences over the mean and variance of the returns on his or her portfolio that are described by the utility function

$$U(\mu_P, \sigma_P) = \mu_P - \left(\frac{A}{2}\right)\sigma_P^2,$$

where A is a parameter that measures the investor's degree of risk aversion. Suppose that this investor is able to firm a portfolio from a risk-free asset with return r_f and a risky asset with expected return equal to μ_r and variance of its return equal to σ_r^2.

a. Letting w denote the share of the investor's wealth allocated to the risky asset and $1-w$ the share allocated to the risk-free asset, write down an equation for the expected return on his or her portfolio.

b. Next, write down an equation for the variance of the return on his or her portfolio.

c. Write down an equation that shows how the investor's optimal choice w^* for the fraction of wealth allocated to the risky asset depends on the risk-free rate r_f, the mean and variance μ_r and σ_r^2 of the return on the risky asset, and the parameter A measuring risk aversion.